Theorem 1.13

Every non-singular matrix can be transformed to an identity matrix, by a sequence of elementary row operations.

As an illustration of the above theorem, let us consider the matrix $A=\begin{bmatrix}2 & -1\\ 3 & 4\end{bmatrix}$.

Then, $|A|=12+3=15\neq 0$. So, A is non-singular. Let us transform A into I_2 by a sequence of elementary row operations. First, we search for a row operation to make a_{11} of A as 1. The elementary row operation needed for this is $R_1\to\left(\frac{1}{2}\right)R_1$. The corresponding elementary matrix is $E_1=\begin{bmatrix}\frac{1}{2} & 0\\ 0 & 1\end{bmatrix}$.

Then, we get $E_1A=\begin{bmatrix}\frac{1}{2} & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}2 & -1\\ 3 & 4\end{bmatrix}=\begin{bmatrix}1 & \frac{-1}{2}\\ 3 & 4\end{bmatrix}$.

Next, let us make all elements below a_{11} of E_1A as 0. There is only one element a_{21}.

The elementary row operation needed for this is $R_2\to R_2+(-3)R_1$.

The corresponding elementary matrix is $E_2=\begin{bmatrix}1 & 0\\ -3 & 1\end{bmatrix}$.

Then, we get $E_2\left(E_1A\right)=\begin{bmatrix}1 & 0\\ -3 & 1\end{bmatrix}\begin{bmatrix}1 & -\frac{1}{2}\\ 3 & 4\end{bmatrix}=\begin{bmatrix}1 & -\frac{1}{2}\\ 0 & \frac{11}{2}\end{bmatrix}$.

Next, let us make a_{22} of $E_2\left(E_1A\right)$ as 1. The elementary row operation needed for this is $R_2\to\left(\frac{2}{11}\right)R_2$.

The corresponding elementary matrix is $E_3=\begin{bmatrix}1 & 0\\ 0 & \frac{2}{11}\end{bmatrix}$.

Then, we get $E_3\left(E_2\left(E_1A\right)\right)=\begin{bmatrix}1 & 0\\ 0 & \frac{2}{11}\end{bmatrix}\begin{bmatrix}1 & -\frac{1}{2}\\ 0 & \frac{11}{2}\end{bmatrix}=\begin{bmatrix}1 & -\frac{1}{2}\\ 0 & 1\end{bmatrix}$.

Finally, let us find an elementary row operation to make a_{12} of $E_3\left(E_2\left(E_1A\right)\right)$ as 0. The elementary row operation needed for this is $R_1\to R_1+\left(\frac{1}{2}\right)R_2$. The corresponding elementary matrix is $E_4=\begin{bmatrix}1 & \frac{1}{2}\\ 0 & 1\end{bmatrix}$.

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